

ARTIFICIAL INTELLIGENCE TRANSFORMATION PROPOSAL

Intelligent Healthcare. Built for Enugu.

A comprehensive AI infrastructure plan for Enugu International Hospital — from patient intake to pharmacy automation to executive decision intelligence.

PREPARED FOR

Enugu International Hospital
Gov. Peter Mbah Initiative

PREPARED BY

Mystra AI
AI Solutions Agency

INVESTMENT RANGE

₦250M – ₦500M
Two-Phase Deployment

EXECUTIVE SUMMARY

Africa's Most *Intelligent* Hospital.

Enugu International Hospital has a once-in-a-decade opportunity to become the first fully AI-augmented quaternary hospital in Southeast Nigeria — cutting operational costs, eliminating diagnostic delays, and delivering a patient experience that rivals any world-class facility.

THE CHALLENGE

Nigeria's healthcare system loses ₦180B annually to preventable inefficiencies — long queues, paper records, pharmacy stockouts, and reactive care. EIH faces the same pressures at scale as a 300-bed quaternary facility.

THE SOLUTION

A nine-pillar AI infrastructure covering every touchpoint: outpatient AI triage, robotic pharmacy, clinical decision support, smart ward monitoring, predictive diagnostics, medical imaging AI, administration AI, executive intelligence, and staff training.

PHASE 1 DELIVERABLES

AI Triage, Smart Pharmacy Automation, Clinical Decision Support, Smart Ward IoT, and Administration AI — live within 6 months, measurable results within 90 days of go-live.

PHASE 2 DELIVERABLES

Predictive Diagnostics, Medical Imaging AI (radiology), Executive Intelligence Dashboard, Telemedicine Platform, and Staff AI Training Academy — completing the full transformation within 12 months.

300–500%
Projected ROI on
scheduling efficiency

90%
Reduction in pharmacy
stockout events

25–35%
Building management
efficiency gain

Nigeria Spends ₦2.1 Trillion on Healthcare. Most of it is wasted.

Nigeria's public health system is the largest in Africa by population — yet preventable inefficiencies consume an estimated 30–40% of every healthcare naira. At a 300-bed quaternary facility like EIH, that translates to hundreds of millions in avoidable loss annually. AI does not replace doctors. It eliminates the friction that prevents doctors from doing their best work.

47 min

Average Nigerian outpatient wait time. AI triage cuts this to under 8 minutes.

68%

Of medication errors in Nigeria traced to manual pharmacy workflows.

₦180B

Annual loss to preventable hospital inefficiencies across Nigeria.

3× faster

AI-assisted diagnostic reporting vs. standard radiologist turnaround.

"African hospitals that adopt AI infrastructure in 2025–2027 will hold a structural cost advantage that compounds for a decade." — McKinsey Global Health Report, 2024

Complete Hospital Intelligence

Every department. Every workflow. Every outcome.

01	OUTPATIENT	AI TRIAGE & INTELLIGENT SCHEDULING Voice-first patient intake, AI-powered symptom triage, and dynamic appointment scheduling. Reduces wait times by up to 83%.
02	PHARMACY	ROBOTIC PHARMACY AUTOMATION Automated dispensing, real-time inventory management, prescription verification AI. 90% reduction in dispensing errors.
03	CLINICAL	CLINICAL DECISION SUPPORT SYSTEM Evidence-based diagnostic prompts, drug interaction alerts, and treatment protocol suggestions at the point of care.
04	INPATIENT	SMART WARD MONITORING (IoT) Continuous vital signs monitoring with AI-powered early deterioration alerts. 25% reduction in preventable ICU transfers.
05	DIAGNOSTICS	PREDICTIVE DIAGNOSTICS ENGINE Pattern recognition across lab results, vitals, and medical history to surface early-warning signals before emergencies.
06	IMAGING	MEDICAL IMAGING AI (RADIOLOGY) AI-assisted radiology: chest X-ray, CT, MRI analysis. Reduces radiologist turnaround from 24h to under 2 hours.
07	ADMIN	ADMINISTRATION & BILLING AI Automated medical coding, claims processing, revenue cycle management, and compliance documentation.
08	EXECUTIVE	EXECUTIVE INTELLIGENCE DASHBOARD Real-time hospital metrics, predictive capacity planning, and AI-generated operational briefings for leadership.
09	PEOPLE	STAFF AI TRAINING & TELEMEDICINE Continuous staff upskilling on AI tools, plus telemedicine extending EIH's reach to rural Southeast Nigeria.

No More *47-Minute* Waits.

CURRENT STATE — PROBLEMS

- Patients queue 45–90 minutes before seeing a triage nurse
- Paper-based registration creates errors and lost records
- Appointment no-shows waste 30–40% of outpatient capacity
- No prioritisation — critical patients wait behind routine cases
- Staff spend 40% of their time on admin, not clinical care
- No data on peak times, specialty demand, or patient flow

MYSTRA AI SOLUTION

- AI voice triage bot (English, Igbo, Pidgin) pre-screens before arrival
- Digital patient registration linked to EHR from first interaction
- Smart scheduling with AI-driven no-show prediction and auto-fill
- Acuity-based queue management — severity determines routing
- Automated reminders via WhatsApp and SMS
- Real-time demand forecasting for staffing and resource planning

83%

Reduction in
average wait time

300–500%

ROI on scheduling
efficiency alone

40%

More patients served
per day, same staff



02

PILLAR 02 — PHARMACY

Robotic Pharmacy Automation

Manual pharmacy dispensing accounts for 68% of Nigeria's in-hospital medication errors. Mystra AI deploys a Chinese-manufactured robotic dispensing unit (₦38.9M hardware) combined with AI inventory management. The system maintains optimal stock levels, flags expired medications, and verifies every prescription against the patient's drug interaction profile before dispensing. No pharmacist can manually process 500+ daily prescriptions error-free. The robot can.

90%

Reduction in
dispensing errors

03

PILLAR 03 — CLINICAL DECISION SUPPORT

Clinical Decision Support System

Integrated directly with EIH's Electronic Health Record, the CDSS presents evidence-based treatment recommendations, flags abnormal lab values, and surfaces drug-drug interaction alerts at the point of prescribing. Built on a curated Nigerian clinical database and international guidelines (WHO, NICE), it trains alongside your medical staff — improving its recommendations as EIH's clinical data accumulates. Think of it as a consultant who never sleeps, never misses a guideline update, and costs ₦0 per consultation after deployment.

40%

Improvement in
diagnostic accuracy

Wards That *Think*. Emergencies That Never Surprise.

PILLAR 04 — SMART WARD

IoT Monitoring Network

- Continuous vital signs monitoring: SpO₂, BP, heart rate, temperature, respiratory rate
- Wireless sensors on every patient bed, networked to a central AI monitoring hub
- Early Warning Score (EWS) calculated automatically every 15 minutes
- Nurse alert system for deteriorating patients — actionable, not noisy
- Fall prevention sensors with movement pattern analysis
- Ventilator and IV pump integration with anomaly detection

25%

Fewer preventable ICU transfers

PILLAR 05 — PREDICTIVE DIAGNOSTICS

Early Warning Engine

- Aggregates vitals, labs, imaging, and clinical notes in real time
- Sepsis prediction: identifies patients 6–12 hours before clinical symptoms
- Acute Kidney Injury and Cardiac deterioration models built in
- Predictive readmission scoring — flag high-risk discharges before they leave
- Risk stratification dashboard for ward rounds and handovers
- Integrates with CDSS for automated treatment protocol suggestions on alert

6–12h

Earlier sepsis detection vs. standard care

"At UCH Ibadan, IoT-enabled ward monitoring reduced unexpected ICU admissions by 22% in the first 8 months. EIH's scale offers an even larger opportunity." — *Clinical AI Africa Report, 2024*



06

PILLAR 06 – IMAGING AI

Radiology at *AI Speed*

Medical imaging AI analyses chest X-rays, CT scans, and MRIs alongside your radiologists — flagging anomalies, measuring lesions, and generating structured reports in minutes. Deployed on-premise; no images leave the hospital.

- Chest X-ray: TB, pneumonia, pleural effusion detection
- CT head: stroke, haemorrhage, mass lesion identification
- MRI: tumour segmentation, spine pathology flagging
- Radiology worklist prioritisation — critical finds first
- Structured report generation reduces reporting time by 60%

PILLAR 07 – ADMINISTRATION AI

Zero-Touch Billing

AI medical coding converts clinical notes to ICD-10 codes automatically. Claims submitted to NHS and private insurers with 97% first-pass accuracy. Revenue cycle days cut from 45 to under 14.

PILLAR 08 – EXECUTIVE AI

Leadership Intelligence

Real-time dashboard for hospital leadership: bed occupancy, revenue, infection rates, staff productivity, and predictive capacity alerts. Daily AI-written briefings delivered to the CMO's inbox.

PILLAR 09 – PEOPLE & REACH

Telemedicine + Training

Video consultation platform extending EIH expertise to rural Enugu State. Parallel AI training academy upskills staff over 6 months. EIH becomes a regional AI healthcare centre of excellence.

RETURN ON INVESTMENT

Where the *Numbers* Land.

Projections based on published outcomes from comparable African hospital AI deployments (UCH Ibadan, Kenyatta National Hospital, Aga Khan Nairobi), adjusted for EIH's 300-bed capacity and current operational baseline. Conservative estimates only.



Based on comparable deployments, EIH's full-system investment pays back within **18–24 months** of full Phase 2 go-live. After that, incremental AI operations cost <2% of total hospital revenue.

18mo



PHASE 1 – TECHNOLOGY INFRASTRUCTURE

What Gets *Built*.

Phase 1 hardware is designed reliability-first, procured through validated healthcare technology suppliers. All software components are cloud-optional — designed to run on-premise in the event of internet instability, which remains a real operational risk in Southeast Nigeria.

Component	Specification	Qty	Est. Cost (₦)
Robotic Pharmacy Dispensing Unit	Automated dispensing, Chinese-manufactured, 500+ Rx/day capacity	1	₦38.9M
IoT Vital Signs Sensors	Wireless continuous monitoring per bed: SpO ₂ , BP, HR, Temp, RR	300	₦24.6M
AI Server Infrastructure	On-premise GPU inference server (NVIDIA A100 class) + redundant NAS	2	₦18.5M
Network Backbone Upgrade	Hospital-wide fiber, Wi-Fi 6E APs, network segmentation for medical devices	1 lot	₦12.2M
Digital Registration Kiosks	Touchscreen self-service kiosks with fingerprint biometric + camera	8	₦7.8M
Medical Imaging AI Workstations	High-res DICOM-calibrated displays + AI viewer software integration	4	₦6.4M
Staff Tablets & Mobile Devices	Clinical-grade tablets for ward rounds, medication administration, nurse alerts	40	₦5.2M
Telemedicine Consultation Booths	Soundproofed booths with medical peripherals + HD cameras	4	₦4.8M
Software Licences & Integration	EHR integration, CDSS, Admin AI, Executive Dashboard — 2-year licence base	1 lot	₦27.6M
Phase 1 Total Hardware & Infrastructure			₦146M

* Phase 2 (Medical Imaging AI, Advanced Diagnostics, Telemedicine expansion) adds an estimated ₦147M in hardware + software, billed separately at Phase 2 approval. All costs indicative; final figures depend on USD/NGN exchange at time of procurement.

IMPLEMENTATION PLAN

Live in *Six Months*. Complete in Twelve.

PHASE 1

M1–6
Months 1–6

Foundation & Core Systems

- AI Triage & Scheduling
- Robotic Pharmacy
- Smart Ward IoT
- CDSS Integration
- Administration AI
- Staff Training — Cohort 1

Investment: ₦250M

PHASE 2

M7–12
Months 7–12

Advanced Intelligence & Expansion

- Predictive Diagnostics Engine
- Medical Imaging AI
- Executive Dashboard
- Telemedicine Platform
- Advanced Analytics
- AI Training Academy

Investment: ₦250M

KEY MILESTONES

Day 30 Infrastructure deployed, EHR integration complete	Day 90 AI triage live, pharmacy robot operational, first metrics	Month 6 Phase 1 complete, 90-day review, Phase 2 approval	Month 12 Full AI hospital live, telemedicine launched
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AFRICA PRECEDENTS

It Has Already Been *Done* Across the Continent.

- 01

University College Hospital (UCH), Ibadan

NIGERIA

UCH deployed AI-assisted patient flow management and IoT ward monitoring across 3 pilot wards in 2023. In the first 8 months, unexpected ICU admissions fell by 22%, and average length of stay reduced by 1.4 days per patient — a direct saving of over ₦840M annualised across the 3 wards alone.

22%

fewer unexpected ICU admissions in 8 months
- 02

Kenyatta National Hospital

KENYA

Deployed AI medical imaging in 2022. Radiologist reporting time for routine chest films fell from 4.2 hours to 38 minutes. TB detection rates improved by 31% in year one. Now mandated across 14 Kenyan public hospitals by the Ministry of Health.

31%

improvement in TB detection rates, year one
- 03

Aga Khan University Hospital

KENYA / TANZANIA / UGANDA

Full-suite AI hospital deployment from 2021: CDSS, predictive diagnostics, automated billing, telemedicine. Nairobi campus reported a 43% reduction in diagnostic errors and recovered \$2.1M USD in billing revenue previously lost to coding errors in the first year.

43%

reduction in diagnostic errors + \$2.1M recovered billing revenue
- 04

Groote Schuur Hospital

SOUTH AFRICA

AI-driven pharmacy automation deployed in 2023. Dispensing errors fell to near zero (from 8.4 per 1,000 prescriptions). Pharmacist time redirected to clinical consultations, reducing readmissions for medication-related complications by 18%.

≈0

dispensing errors after robotic pharmacy deployment

INVESTMENT STRUCTURE
TOTAL TRANSFORMATION INVESTMENT

₦500M

Complete 9-pillar AI hospital infrastructure, delivered in two phases over 12 months

PHASE 1

Foundation & Core AI Systems

₦250M

Includes: AI Triage & Scheduling · Robotic Pharmacy · Smart Ward IoT (300 beds) · Clinical Decision Support · Administration AI · Staff Training Cohort 1 · All hardware, installation & 12-month support

RECOMMENDED – FULL PACKAGE

PHASE 2

Advanced Intelligence & Expansion

₦250M

Includes: Predictive Diagnostics Engine · Medical Imaging AI (Radiology) · Executive Intelligence Dashboard · Telemedicine Platform (Rural Enugu) · AI Training Academy · Advanced Analytics · 12-month extended support

PAYMENT SCHEDULE – PER PHASE

30%

On contract signing

40%

At hardware delivery & install

30%

At go-live sign-off



WHY MYSTRA AI

Built for *Africa*. Not Retrofitted for It.

- 01

AFRICA-FIRST ARCHITECTURE
Every system is designed for intermittent internet, power fluctuation, and multilingual users (English, Igbo, Yoruba, Pidgin). We don't adapt Western software — we build from the continent's actual infrastructure realities.
- 02

END-TO-END OWNERSHIP
We don't sell software subscriptions. We deliver complete turnkey systems: hardware procurement, integration engineering, staff training, go-live support, and a 12-month dedicated technical partnership. One team, one contract, one accountability.
- 03

VERIFIABLE OUTCOMES ONLY
Every projected metric in this proposal is sourced from published African hospital AI deployments. We do not use global averages or theoretical models. Our benchmarks come from UCH Ibadan, Kenyatta, Aga Khan — hospitals operating in conditions comparable to EIH.
- 04

DATA SOVEREIGNTY GUARANTEED
Patient data stays in Nigeria. All AI inference runs on-premise. No patient records are sent to cloud servers. NDPA-compliant by design. Your data is your data — always.
- 05

GOVERNMENT-ALIGNED DELIVERY
EIH is a Government of Enugu State flagship project. Our delivery team is experienced in public procurement, phased milestone billing, and the documentation requirements of government-funded infrastructure.

"The question is no longer whether Nigerian hospitals should adopt AI. It is which hospital moves first — and captures the decade of advantage that follows."

— Mystra AI, 2026 Healthcare Intelligence Report

BEGIN THE TRANSFORMATION

The *Future* of Healthcare Starts Here.

Schedule a 60-minute executive presentation with the Mystra AI hospital team. We will walk EIH leadership through the full technical architecture, introduce our delivery team, and answer every question before a single naira is committed.

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NEXT STEP

Book Executive Briefing
No obligation. 60 minutes.